HACKCULT '26 · HC26-GEDA01 · GREEN ENERGY & SUSTAINABILITY

LiFE-OS

Your electricity bill, turned into a footprint, savings, a solar plan and community action.

An interactive Mission LiFE citizen engagement application for the Gujarat Energy Development Agency.
Next.js · TypeScript · Supabase PostgreSQL

1 · THE PROBLEM

Awareness is high. Daily action is not.

Friction

Carbon calculators ask dozens of questions, so most people quit before seeing a number.

No link to rupees

Households respond to a lower bill. Kilograms of CO_2 alone rarely change a habit.

No path to rooftop solar

Interest stalls at "is it worth it for my roof?" with no honest estimate and no clear next step.

Who it serves: residential consumers, local communities, and state administrators monitoring regional conservation.

2 · THE SOLUTION

All four required features, working end to end

Personalised footprint tracker

One-minute onboarding from last month's units. Baseline, monthly trend, travel and LPG add-ons, and a visible calculation.

Nudges and micro-tasks

Suggestions from the user's own bills and the time of day, each explaining why. Ten daily, weekly and one-time tasks.

Gamification and social impact

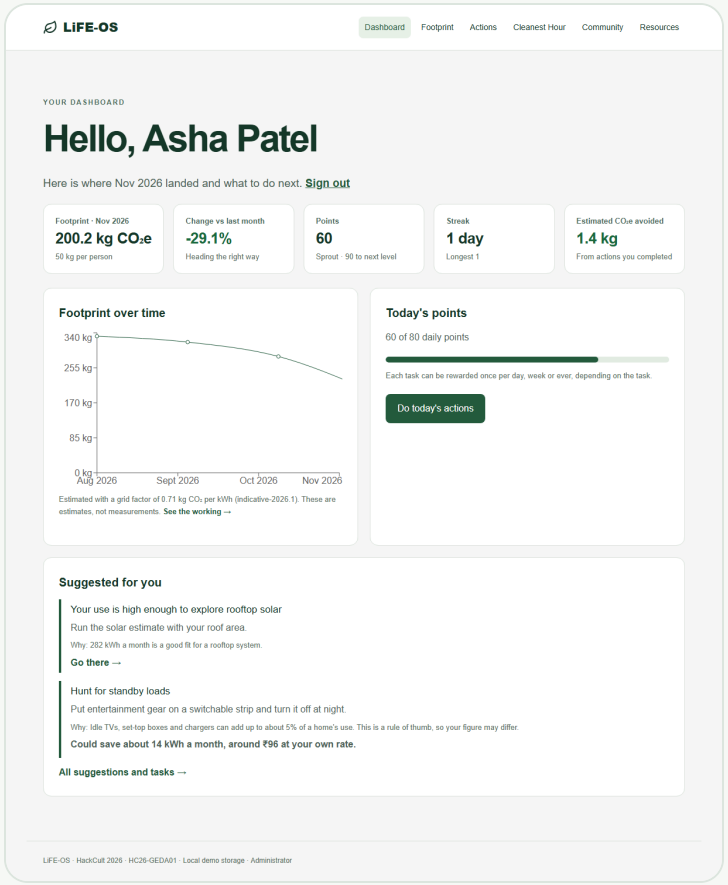
Points, streaks, levels, local and state-wide challenges with shared targets, and opt-in leaderboards.

Actionable resource hub

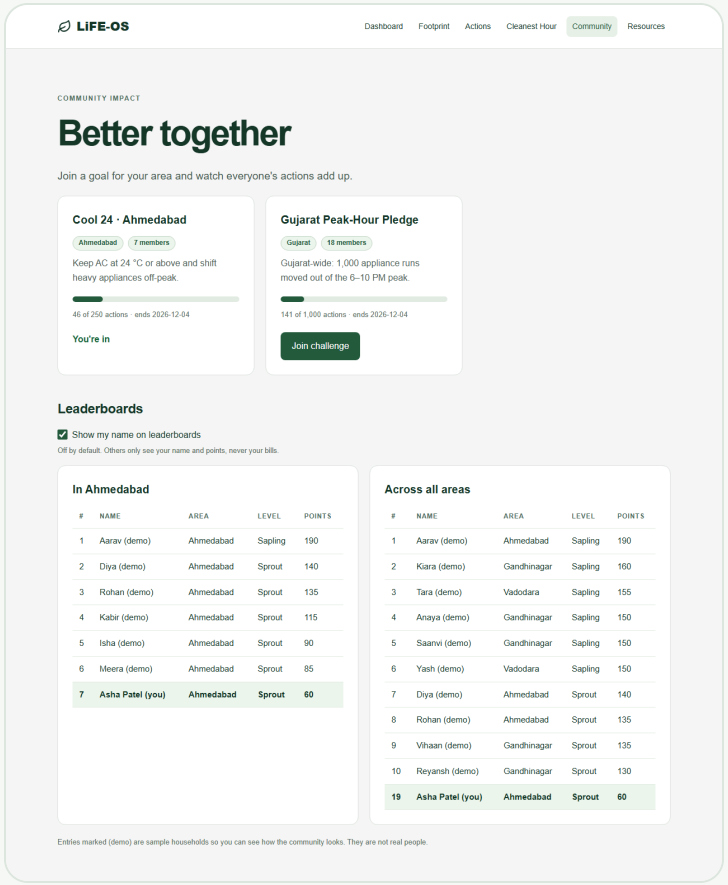
Rooftop solar estimate with ranges and assumptions, the steps to apply, and verified links to official bodies.

Plus: area-level admin view for GEDA with privacy suppression, and "Cleanest Hour" appliance scheduling.

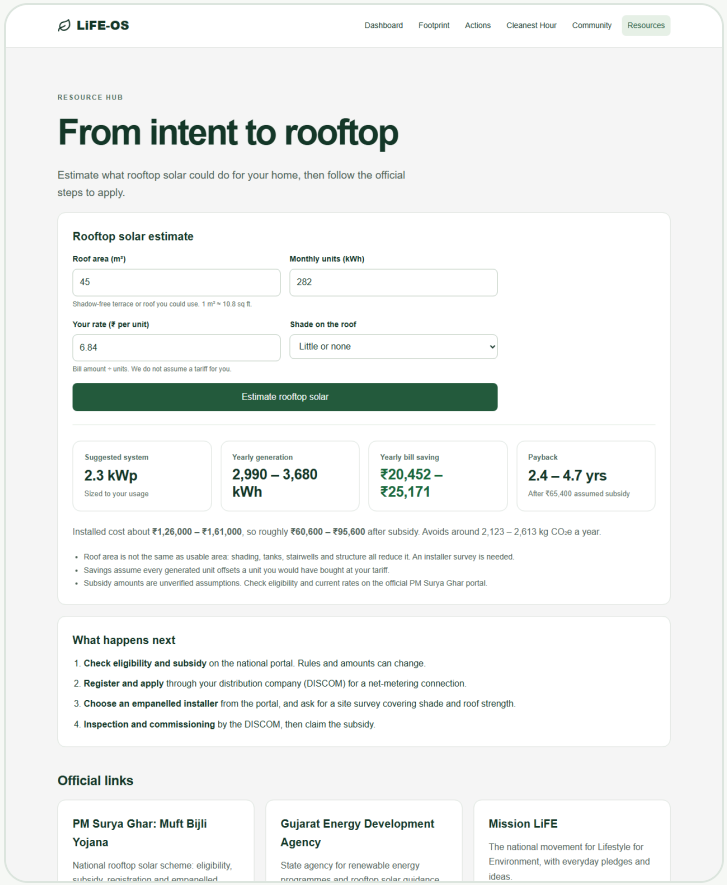
Real screens, real data flow



Dashboard: footprint, trend, points, nudges



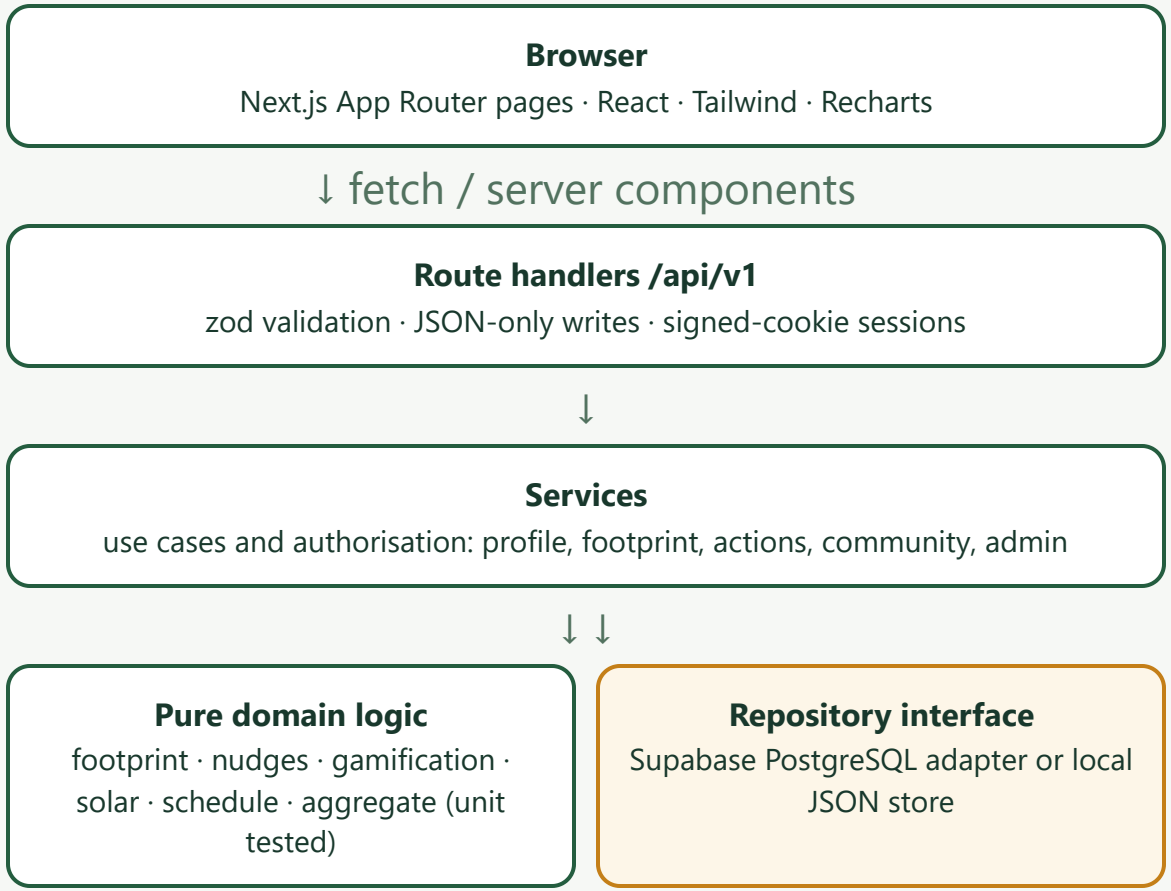
Community: challenges and leaderboards



Resource hub: solar estimate with assumptions

4 · ARCHITECTURE

Layered, with one persistence boundary



Why it is built this way

- Business rules are pure functions, so they are testable and portable.
- Swapping the database touches one interface, not the app.
- Authorisation lives on the server, never in a hidden menu link.
- Zero-setup demo mode: the same code runs without any cloud account.

5 · DATABASE SCHEMA

Five tables, constraints that enforce the rules

TABLE	KEY COLUMNS
app_users	id, pincode, household_size, leaderboard_opt_in, is_demo
footprint_snapshots	user_id, period, electricity_kwh, bill_amount_inr, total_kg, factor_version
task_completions	user_id, task_id, dedupe_key, completed_on, points, impact_kg
challenges	region_prefix, metric, target, starts_on, ends_on
challenge_members	challenge_id, user_id

Integrity in the database

- **unique(user_id, dedupe_key)** makes a double reward impossible, even under concurrent requests.
- **unique(user_id, period)**: saving a month again replaces it.
- **factor_version** on every snapshot keeps old estimates reproducible when factors change.
- **Row level security on, no public policy.** The publishable key can read nothing; only the server can.

6 · TECH STACK AND DECISIONS

One deployable app, no glue services

Next.js 16 · React 19 · TypeScript strict · Tailwind 4 · Recharts · zod · Supabase PostgreSQL · Vitest

25 unit tests cover footprint maths, nudges, solar sizing, points rules, streaks, challenge progress, admin suppression, scheduling and IST date handling. Lint, typecheck and production build all pass.

Key decisions

- Next.js route handlers instead of a separate API server: one deploy, one language.
- Server-signed anonymous sessions keep onboarding to a minute. Supabase Auth is the next step.
- India Standard Time for every "today", so streaks and limits match users' days.

7 · WHY IT STANDS OUT: TRUST BY DESIGN

Honest numbers, fair play, private by default

Rupees from the user's own bill

Savings use their bill ÷ units. We never guess a tariff.

Explain every nudge

Each suggestion states why it fired and how its saving was estimated.

Fair points

One reward per window, a daily cap, and server-side scoring. The client never sends points.

Privacy

Leaderboards are opt-in. Admins see only areas with 5+ households, grouped by 3 PIN digits.

Every estimate shows its assumptions, and unverified inputs are labelled as such in the interface.

Built to grow, honest about today

Scales because

- Stateless, pure domain logic scales horizontally on serverless.
- Idempotent writes enforced by database constraints.
- Indexes on per-user history and PIN-prefix grouping.
- Next: admin and leaderboards move to SQL views (`GROUP BY left(pincode,3) HAVING count(*) >= 5`) with paginated ranking.

Limitations today

- Emission factors are indicative, not the current CEA release.
- Solar subsidy figures are unverified assumptions.
- Cleanest Hour is a modelled curve, not live grid data.
- Tasks are self-reported. Sessions are anonymous.
- Supabase adapter not yet run against a live project.
- No bill OCR, roof tracing or AI assistant yet.

From prototype to statewide programme

Next

- Supabase Auth (phone OTP) with per-user RLS
- Bill upload with OCR and a mandatory review step
- Roof outline on a map with PVGIS yield

Then

- DISCOM tariff slabs from official GERC orders
- Hourly grid intensity series for Cleanest Hour
- Gujarati and Hindi interfaces

The ask

Pilot LiFE-OS with a DISCOM and a few wards: verified tariffs, real bill data, and measured savings.